Toric Varieties

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Abstract

Contents

1 Combinatorial Commutative Algebra

1.1

Definition 1.1.1 (Monomial Notation)
Let R be a commutative ring. Let $u = (u_1, \dots, u_n) \in \mathbb{N}^n$. Define

$$x^u = x_1^{u_1} \cdots x_n^{u_n} \in R[x_1, \dots, x_n]$$

1.2 The Semi-group Ring

Definition 1.2.1 (The Semi-Group Ring)
Let S be a monoid. Let R be a ring. Define the semi-group ring of S over R to be the left R-module

$$R[S] = \bigoplus_{s \in S} R \cdot s$$

together with multiplication defined on basis by $t^{s_1} \cdot t^{s_2} = t^{s_1 + s_2}$

Lemma 1.2.2
Let S be a monoid. Let R be a ring. Then the following are true.
- $R[S]$ is a ring.
- If S is commutative, then $R[S]$ is a commutative ring.
- If R and S are commutative, then $R[S]$ is an R-algebra.

1.3 Polyhedral Cones

Definition 1.3.1 (The Cone of a Set of Vectors)
Let k be a field. Let V be a finite dimensional vector space over k. Let $S = \{v_1, \dots, v_s\} \subseteq V$ be finite set of vectors. Define the cone spanned by S to be

$$\text{Cone}(S) = \left\{ \sum_{k=1}^{s} \lambda_k v_k \;\middle|\; \lambda_k \geq 0 \right\}$$

Definition 1.3.2 (Cones in a Vector Space)
Let k be a field. Let V be a finite dimensional vector space over k. Let $C \subseteq V$ be a subset of V. We say that C is a cone of V if there exists a finite subset $S \subseteq V$ of vectors such that

$$C = \text{Cone}(S)$$

Lemma 1.3.3
Let k be a field. Let $C \subseteq k^n$ be a subset of V. Then C is a cone if and only if there exists half planes $H_1, \dots, H_d \subseteq k^n$ such that

$$C = H_1 \cap \cdots \cap H_d$$

Definition 1.3.4 (Dual Cone)
Let k be a field. Let V be a finite dimensional vector space over k. Let $C \subseteq V$ be a subset of V. Then define the dual cone of C to be

$$C^\vee = \{f \in V^* \mid \langle f, v \rangle \geq 0 \text{ for all } v \in C\}$$

Lemma 1.3.5
Let

1.4 Affine Semi-groups

Definition 1.4.1 (Affine Semi-group)
Let S be a monoid. We say that S is an affine semi-group if there exists an injective monoid homomorphism $S \to \mathbb{N}^d$ for some $d \in \mathbb{N}$.

1.5 Lattice Ideals

Let S be a commutative monoid. Write $G(S)$ the grothendieck completion of S. It is an abelian group.

Definition 1.5.1 (Lattice Ideal)
Let $L \subseteq \mathbb{Z}^n$ be a lattice. Let R be a commutative ring. Define the lattice ideal of L in $R[x_1, \dots, x_n]$ to be

$$I_L = R\langle x^u - x^v \mid u, v \in \mathbb{N}^n \text{ and } u - v \in L\rangle$$

Proposition 1.5.2
Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S. Let $\phi : \mathbb{Z}^n \to G(S)$ be the group homomorphism defined by $e_i \mapsto s_i$. Then there is a k-algebra isomorphism

$$k[S] \cong \frac{k[x_1, \dots, x_n]}{I_{\ker(\phi)}}$$

Proposition 1.5.3
Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S. Let $\phi : \mathbb{Z}^n \to G(S)$ be the group homomorphism defined by $e_i \mapsto s_i$. Then the following are equivalent.

- $I_{\ker(\phi)}$ is prime.
- $k[S]$ is an integral domain.
- $G(S)$ is a free abelian group.
- There exists an injective monoid homomorphism $S \to \mathbb{N}^d$ for some $d \in \mathbb{N}$.

2 Constructing Toric Varieties

2.1 The Standard n-Torus

Definition 2.1.1 (The Standard n-Torus)
Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Define the standard n-torus over k to be the affine group variety

$$\mathbb{G}_m^n = (\mathbb{C}^*)^n$$

Definition 2.1.2 (Tori)
Let k be a field. Let V be an affine variety over k. We say that V is a torus if there exists $n \in \mathbb{N} \setminus \{0\}$ such that there is a isomorphism of varieties

$$V \cong \mathbb{G}_m^n$$

Recall that the functor $D : \mathbf{FGAb} \to \mathbf{Diag}$ sending a finitely generated abelian group G to $D(G) = \text{Spec}(k[G])$ is an equivalence of categories. Evidently tori corresponds to the free abelian groups of finite rank under the equivalence.

Lemma 2.1.3
Let k be a field. Let T be torus over k. Then the following are true.
- Let T' be a torus over k. Let $\phi : T \to T'$ be a morphism of group schemes. Then $\text{im}(\phi)$ is a torus.
- Let $H \leq T$ be a subgroup variety of T. If H is irreducible, then H is a torus.
- Let $H \leq T$ be a subgroup of T. Then T/H is a torus.

Definition 2.1.4 (Character Lattices)
Let k be a field. Let T be a torus over k. Define the character lattice of T to be the group of characters $\chi(T)$.

Since χ is the inverse to D, we have the following result.

Lemma 2.1.5
Let k be a field. Let $T \cong \mathbb{G}_m^n$ be a torus over k. Then there is an isomorphism

$$\chi(T) \cong \mathbb{Z}^n$$

given as follows. For each $m \in \mathbb{Z}^n$, the corresponding character is the map $\chi^m : T \to \mathbb{G}_m$ defined by $\chi^m(t) = t^m$

Definition 2.1.6 (One-parameter Subgroups)
Let k be a field. Let T be a torus over k. A one-parameter subgroup of T is a homomorphism of group schemes $\mathbb{G}_m \to T$.

Lemma 2.1.7
Let k be a field. Let $T \cong \mathbb{G}_m^n$ be a torus over k. Then there is an isomorphism

$$\text{Hom}_{\mathbf{GrpSch}_k}(\mathbb{G}_m, T) \cong \mathbb{Z}^n$$

Lemma 2.1.8
Let k be a field. Let T be a torus over k. Then the map

$$\chi(T) \times \text{Hom}_{\mathbf{GrpSch}_k}(\mathbb{G}_m, T) \to \mathbb{Z}$$

defined by $(\chi, \lambda) \to \chi \circ \lambda$ is a perfect pairing.

2.2 Constructing Toric Varieties

Definition 2.2.1 (Toric Varieties)
Let k be a field. Let X be an irreducible variety. We say that X is a toric variety if the following are true.

- $(\mathbb{C}^*)^n$ is a Zariski open subset of X.
- The action of $(\mathbb{C}^*)^n$ on itself extends to X.

We denote the underlying torus of X by T_X.

Definition 2.2.2 (Affine Toric Varieties Associated to Characters)
Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Let $\mathcal{A} = \{\chi_1, \dots, \chi_m\} \subset \chi(\mathbb{G}_m^n)$ be characters of the standard n-torus. Define the affine toric varieties associated to $\mathcal{A}$ to be

$$Y_{\mathcal{A}} = \overline{\text{im}(\Phi_{\mathcal{A}})}$$

where $\Phi_{\mathcal{A}} : \mathbb{G}_m^n \to \mathbb{A}_{\mathbb{C}}^m$ is the map defined by $\Phi_{\mathcal{A}}(t) = (\chi_1(t), \dots, \chi_m(t))$

Proposition 2.2.3
Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Let $\mathcal{A} \subset \chi(\mathbb{G}_m^n)$ be a finite subseet of characters of the standard n-torus. Then the following are true.

- $Y_{\mathcal{A}}$ is an affine toric variety.
- The underlying torus of $Y_{\mathcal{A}}$ has character lattice $\mathbb{Z}\mathcal{A}$.
- $\dim(Y_{\mathcal{A}}) = \text{rank}(\mathbb{Z}\mathcal{A})$

Definition 2.2.4 (Toric Ideals)
Let $L \subseteq \mathbb{Z}^n$ be a lattice. Let R be a commutative ring. Let $I_L \subseteq R[x_1, \dots, x_n]$ be the lattice ideal of L. We say that I_L is a toric ideal if I_L is a prime ideal.

Proposition 2.2.5
Let k be a field. Let X be an irreducible affine variety over k. Then the following are equivalent.

- X is a affine toric variety.
- There exists a lattice L and a finite subset $\mathcal{A} \subseteq L$ such that $X = Y_{\mathcal{A}}$.
- There exists a toric ideal I of $k[x_1, \dots, x_n]$ such that $X = \text{Spec}\left(\frac{k[x_1,\dots,x_n]}{I}\right)$.
- There exists an affine semigroup S such that $X = \text{Spec}(k[S])$.

Proposition 2.2.6
Let k be an algebraically closed field. Let S be an affine semi-group. Then there is a bijection

$$\{p \in \text{Spec}(k[S]) \mid p \text{ is a closed point }\} \overset{1:1}{\longleftrightarrow} \{\phi : S \to (\mathbb{C}, \cdot) \mid \phi \text{ is a monoid homomorphism }\}$$

given as follows. For each $p \in \text{Spec}(k[S])$ a closed point, p corresponds to the monoid homomorphism $\gamma_p : S \to \mathbb{C}$ defined by $\gamma_p(m) = \chi^m(p)$

Proposition 2.2.7
Let k be an algebraically closed field. Let $X = \text{Spec}(k[S])$ be an affine torus over k. Then the action of the torus T_X on X is induced by the k-algebra homomorphism

$$k[S] \to k[\chi(T_X)] \otimes_k k[S]$$

given by $\gamma_p \mapsto \gamma_p \otimes \gamma_p$ where here we identify the closed points on $k[S]$ by the above prp.

2.3 Normality Condition and Polyhedral Cones

Definition 2.3.1
Let $\sigma \subseteq \mathbb{R}^n$ be a strongly convex rational polyhedral cone. Define the affine toric variety associated to σ to be
$$V_\sigma = \text{Spec}(\mathbb{C}[S_\sigma])$$

Proposition 2.3.2
Let k be a field. Let T be an affine torus variety over k. Then the following are equivalent.
- X is normal.
- There exists a saturated affine semigroup S such that $X = \text{Spec}(\mathbb{C}[S])$.
- There exists a strongly convex rational polyhedral cone σ such that $X = \text{Spec}(\mathbb{C}[S_\sigma])$.

2.4 Criteria for Smoothness